

Exercise 1

SELECT & FILTERING

Selecting & Sorting

1.

SELECT *

FROM employees;

id	first_name	Last-Name	department	Salary	hire_date	City
1	Alice	Green	IT	70 000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45 000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65 000	2018-07-18	Durban
4	David	MakGONA	Marketing	50 000	2021-11-05	Pretoria
5	Eva	Maidoo	IT	72 000	2017-09-30	Johannesburg

2.

1) SELECT DISTINCT department

FROM employees;

department
IT
HR
FINANCE
MARKETING

3.

```
SELECT First-Name,
        Last-Name
FROM    employees
ORDER BY Salary DESC;
```

First-Name	Last-Name
Eva	Naidoo
Alice	Green
Carthy	Zulu
David	Mokdona
Brian	Lee

4.

```
SELECT Id,
        First-Name,
        Last-Name,
        Salary
FROM    employees
ORDER BY Salary DESC
LIMIT 3;
```

Id	First-Name	Last-Name	Salary
5	Eva	Naidoo	72 000
1	Alice	Green	70 000
3	Carthy	Zulu	60 000

Filtering with WHERE

5.

```
SELECT id,  
       first-name,  
       last-name,  
       department  
FROM   employees  
WHERE  department = 'IT';
```

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Maidoo	IT

6.

```
SELECT id,  
       first-name,  
       last-name,  
       department,  
       salary  
FROM   employees  
WHERE  department = 'Finance' AND salary > 60000;
```

id	first-name	last-name	department	salary
3	Cathy	Zulo	Finance	65 000

7.

```

SELECT id,
       first-name,
       last-name,
       department
FROM   employees
WHERE  department = 'HR' OR department = 'Marketing';

```

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	MUKOENA	Marketing

8.

```

SELECT id,
       first-name,
       last-name,
       department
FROM   employees
WHERE  department != 'IT';

```

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	MUKOENA	Marketing

9.

```
SELECT id,
       first_name,
       last_name,
       department
FROM   employees
WHERE  department IN ('IT', 'HR', 'Finance');
```

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	FINANCE
5	Eva	Naidoo	IT

10.

```
SELECT id,
       first_name,
       last_name,
       department,
       salary,
       city
FROM   employees
WHERE  department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
```

id	first_name	last_name	department	salary	city
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg